



PPBNG Operation Manual

Internal operator guide · Latest update by [Yiyuan Lin](#) on Oct 03, 2026

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Note

Remote Desktop Connection is a remote desktop software on the Windows laptop for remote control.

HyperFusion is a software on the PPBNG PC for hyper spectral cameras control and visualization.

0. Remote in the PPBNG PC

Connect to the robot network `RUT_XXXX`. Then open **Remote Desktop Connection**. Connect to 192.168.5.150` (PPBNG PC).

1. Set exposure

Open **HyperFusion**. Adjust FX10e exposure, then close **HyperFusion**.

Open this file (pinned to quick access in the File Explorer):

```
C:\Users\cairlab\Desktop\Projects\PPB_NG_2026\ppbng_ros2_ws\src\ppbng_bringup\config\ppbng_config.yaml
```

Edit line 165, `hsi → fx10e → exposure_us`, and save. YAML value = **HyperFusion** exposure in ms × 1000.

Example: **4.5 ms** → **4500.0**. Line number checked September 26, 2026; locate the key if lines change.

2. Open three terminals (PowerShell)

Right click the Windows Start icon → Windows PowerShell. Open three windows.

Warning

Do **NOT** open the Windows PowerShell (Admin)

3. PowerShell 1: Raspberry Pi basic bringup

```
ssh cairlab@192.168.5.200
```

After login, run:

```
ros2 launch amiga_navigation basic_bringup.launch.py
```

Leave it running. Confirm RTK Fixed and the thermal safety monitor is running.

4. PowerShell 2: Windows acquisition

Run this single command:

```
"C:\Users\cairlab\Desktop\Projects\PPB_NG_2026\ppbng_ros2_ws\tools\run_production_acquisition.cmd" trams_20261001_row1 60
```

Tip

Change `trams_20261001_row1` to your dataset name. 60 is the maximum recording duration in minutes.

Important

1. At the cover prompt: cover both HSI lenses, then press `Enter` **once**.
2. After dark capture: remove both covers, then press `Enter` **once**.
3. Wait for `Full acquisition started`.

5. PowerShell 3: start waypoint follower

```
ssh cairlab@192.168.5.200
```

After login, wait for `Full acquisition started` in PowerShell 2, then run:

```
ros2 run amiga_navigation waypoint_follower --waypoints  
/home/cairlab/navigation_waypoints/traminette_row1.csv
```

Tip

Change `traminette_row1.csv` to your waypoint file name.

6. Stop

Press `Enter` (recommended) or `CTRL+C` **once** in PowerShell 2 to stop data acquisition. Robot will detect the sensor closure and stop navigation. Then press `CTRL+C` in PowerShell 1 and PowerShell 3. Put the covers back to the camera lenses.